

PTIS Test Result

Premier Tubular Inspection Services

Employee Information

Employee ID: 1589

Name: Mr Zain Ali

Standard: MPT (General)

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Status: Pass

Score: 79.17%

Total Questions: 30

Pass Criteria: 75%

Test Summary

[OK] Correct: 25



[X] Wrong: 5



Detailed Answers

Q1. The Pie Gage Is So Named Because:

A. It is made from soft steel.

[OK] B. It is made of Pie – Shaped sections.

C. The reading is mushy and not predictable.

D. It is round and flat.

[OK] CORRECT

Q2. Demagnetisation with reversing DC is more effective than AC because:

[OK] A. DC is more penetrating

B. Demagnetisation is assisted by the skin effect

C. DC is more direct

D. Not true - AC is more effective

[OK] CORRECT

Q3. Which of the following particles would be most sensitive?

A. Wet

[OK] B. Dry

C. Depends on the test piece permeability

D. None of the above

[OK] CORRECT

Q4. The primary reason for using water rather than oil as a suspension medium for wet method baths is that:

[X] A. Water is more chemically inert than oil

[OK] B. Bath flammability hazards are eliminated

C. Water has the capability to dissolve the needed rust inhibitors

D. Water baths may be used at lower temperatures than oil baths

[X] WRONG - Correct Answer: B

Q5. Excessively High Amperage Is Desirable To Detect The Minute Flaws:

A. TRUE

[OK] B. FALSE

[OK] CORRECT

Q6. Which of the following describes the shape of particles used for dry magnetic particle testing?

A. Spherical

B. Angular

C. Elongated

[OK] D. Mixture of elongated and globular

[OK] CORRECT

Q7. The best method of inducing a circular field in a tube is by a:

[OK] A. Central conductor

B. Head shot

C. Coil

D. Prod technique

[OK] CORRECT

Q8. Which of the following is an advantage of the wet method?

A. Excellent detection of completely subsurface discontinuities

[OK] B. Ease of bath recovery and re-use

C. Low flash point ensures freedom from fire hazards

D. Relatively clean and easy to work with

[OK] CORRECT

Q9. The temperature above which most soft steels become nonmagnetic is about:

A. 440°C (770°F)

B. 523°C (975°F)

[X] C. 626°C (1160°F)

[OK] D. 754°C (1390°F)

[X] WRONG - Correct Answer: D

Q10. The Ability Of A Material To Remain Magnetic After The Magnetising Force Is Removed Is Called:

A. Reluctance

[OK] B. Retentivity

C. Permeability

D. Electromagnetism

[OK] CORRECT

Q11. The Materials Weekly Attracted By A Magnet Are Called:

A. Diamagnetic materials.

[OK] B. Paramagnetic materials.

C. Ferromagnetic materials.

D. Austenitic materials.

[OK] CORRECT

Q12. In Which Magnetizing Method Is The Current Passed Directly Through The Part, Thereby Setting Up A Magnetic Field At Right Angles To The Current Flow?

A. Longitudinal magnetization

B. Coil magnetization

[X] C. Central conductor magnetization

[OK] D. None of the above

[X] WRONG - Correct Answer: D

Q13. How can parts be tested to determine if they have been adequately demagnetized?

A. By bringing a suspended paper clip near the middle of the part

B. By using a small horseshoe permanent magnet

[OK] C. By using a small magnetometer held at a corner of the part

D. By sprinkling some magnetic particles on the part

[OK] CORRECT

Q14. The Important Difference Between AC And DC Current For Magnetic Particle Testing Purposes Is:

A. The skin effect caused by DC adds mobility to the magnetic particles

B. The resulting AC magnetic fields are more difficult to demagnetize

[OK] C. The DC magnetic fields are more penetrating

D. The AC magnetic fields are stronger

[OK] CORRECT

Q15. For a 7.6 cm (3 inches) diameter bar how much current is needed to magnetize the bar for the detection of longitudinal discontinuities:

A. 5500 amperes

B. 16500 amperes

C. 1000 amperes

[OK] D. 3000 amperes

[OK] CORRECT

Q16. Which of the following are ferromagnetic materials?

A. Aluminium, iron, copper

[X] B. Iron, copper, nickel

C. Copper, aluminium, silver

[OK] D. Iron, cobalt, nickel

[X] WRONG - Correct Answer: D

Q17. The 'skin' effect would be most noticeable in which of the following?

A. A magnetic conductor carrying a DC current

B. A nonmagnetic conductor carrying a DC current

C. A magnetic conductor carrying a 50 Hz AC current

[OK] D. A nonmagnetic conductor carrying a 50 Hz AC current

[OK] CORRECT

Q18. The magnetic field outside a conductor decreases:

- A. Exponentially
- [OK] B. In a linear manner**
- C. Inversely with distance
- D. Inversely with the square of distance

[OK] CORRECT

Q19. An inspection for surface and subsurface discontinuities in ferromagnetic welds would best be accomplished by which of the following techniques?

- [OK] A. Prods**
- B. AC yoke
- C. Half wave DC yoke
- D. None of the above

[OK] CORRECT

Q20. Magnetic Particles Testing Is Most Likely To Find Subsurface Discontinuities In:

- [OK] A. Soft steels with high permeability**
- B. Soft steels with low permeability
- C. Hardened steels with low permeability
- D. Hardened steels with high permeability

[OK] CORRECT

Q21. For direct contact magnetising methods, the magnetic field is oriented in what direction relative to the current direction?

- A. Parallel
- [X] B. At 45 degrees**
- [OK] C. At 90 degrees**
- D. At 180 degrees

[X] WRONG - Correct Answer: C

Q22. An Oil Suspension With A High Flashpoint Is Desirable To Provide:

- [OK] A. Low flammability**
- B. High Flammability.
- C. Good mobility.
- D. A Pleasant Odor.

[OK] CORRECT

Q23. Continuous magnetization provides the most sensitivity because:

- A. The magnetic particles are present while the part is being magnetized
- B. The magnetic field is greatest while the magnetizing current is on
- [OK] C. All of the above**
- D. Neither of the above

[OK] CORRECT

Q24. A group of indications, some sharp and some broad and fuzzy, were found on an area of a small forging. Demagnetization and re-inspection eliminated these indications. What was the probable cause?

- A. Forging lap
- [OK] B. Magnetic writing**
- C. Change in permeability
- D. Subsurface variation

[OK] CORRECT

Q25. Magnetic lines of force enter and leave a magnet at:

- A. Saturation
- B. L/D ratios of greater than 4 to 1
- C. Flux concentration points

[OK] D. Poles

[OK] CORRECT

Q26. The most common harmful effect of exposure to black light is:

[OK] A. Eyeball fluorescence

- B. Burned retinas
- C. Skin burns
- D. None of the above

[OK] CORRECT

Q27. An internal rupture caused by working steel at improper temperatures is called a:

- A. Lap
- B. Cold shut

[OK] C. Forging burst

- D. Slag inclusion

[OK] CORRECT

Q28. When using prods, arc burns may be caused by which of the following?

- A. Dirty contact tips
- B. Inadequate pressure
- C. Too large a magnetic current

[OK] D. All of the above

[OK] CORRECT

Q29. Magnetic flux density is zero at:

- A. The inside surface of a tube magnetised with a central conductor
- B. The outside surface of a tube magnetised with a central conductor
- C. The outside surface of a bar magnetised with a head shot

[OK] D. The centre of a bar magnetised with a head shot

[OK] CORRECT

Q30. The Materials Strongly Attracted By A Magnet Are Called:

- A. Diamagnetic materials.
- B. Paramagnetic materials.

[OK] C. Ferromagnetic materials.

- D. Austenitic materials.

[OK] CORRECT
